

## Reminders

### Upcoming due dates

Wed Feb 18 Data checkpoint

Fri Feb 20 D6

# Machine Learning

## Data Science in Practice

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Dept. of Cognitive Science

UC San Diego

A survey!



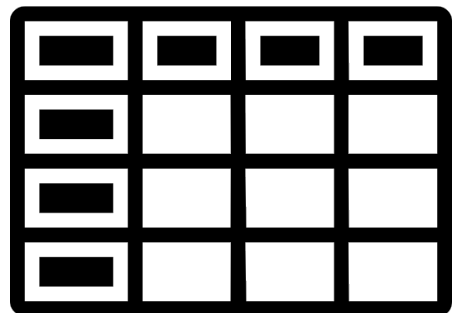
- **Problem:** Detecting whether credit card charges are fraudulent.
- **Data science question:** Can we use the time of the charge, the location of the charge, and the price of the charge to predict whether that charge is fraudulent or not?
- **Type of analysis:** Predictive analysis



Robert Hecht-Neilsen and Zeus (and others) sold HNC for \$810M in 2002  
Around here lots of people see him as a major contributor to the  
development of neural networks and data science as we know them today

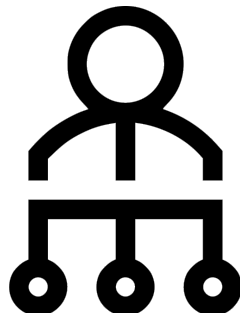
**predictive analysis**  
uses data you have now  
to make predictions in  
the future

**machine learning**  
approaches are used for  
predictive analysis!



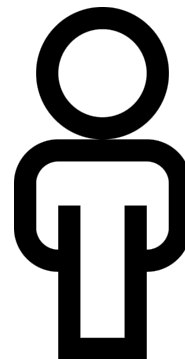
data

train →

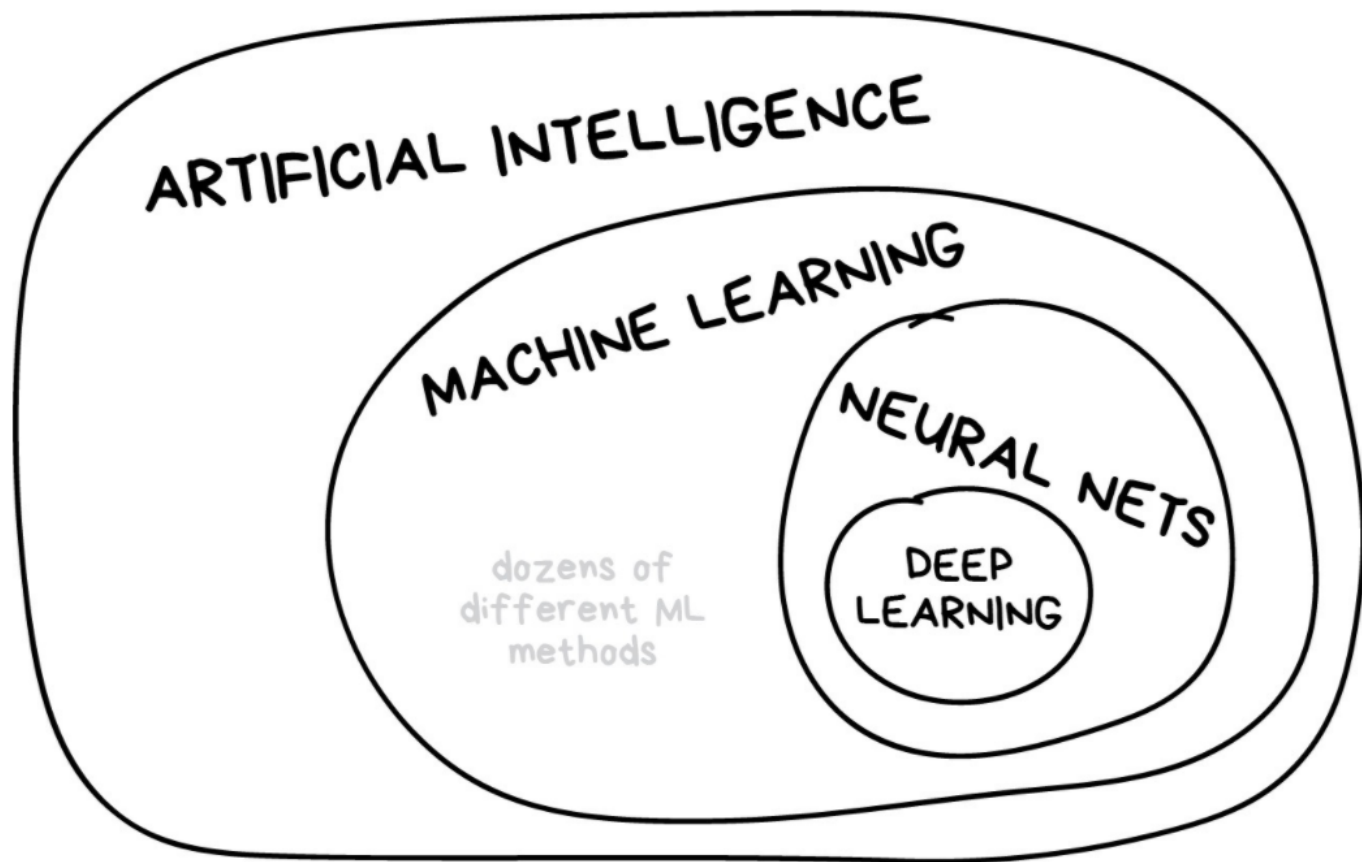


model

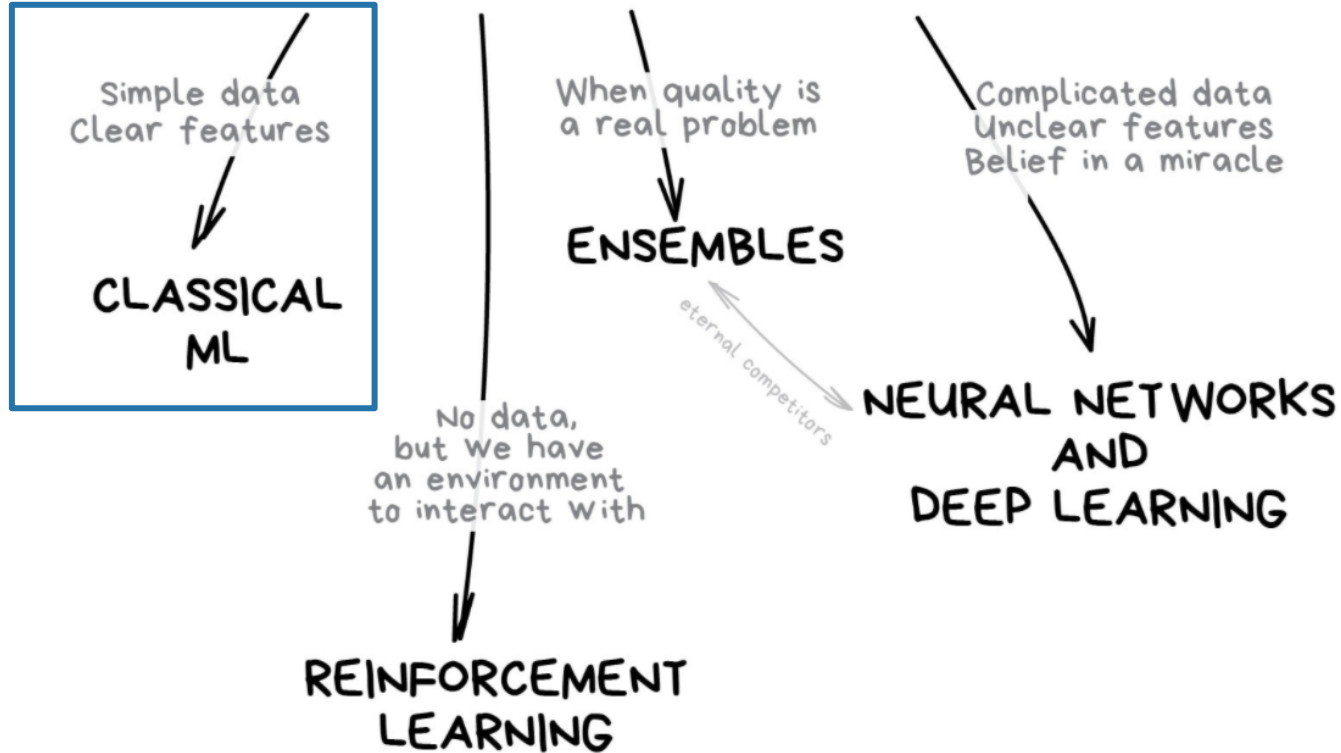
→ predict



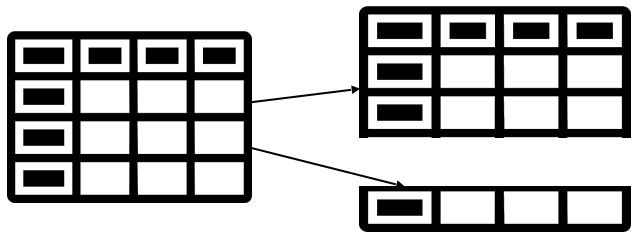
In contrast to statistical approaches which care more about the model accurately reflecting the process than nailing the predictions



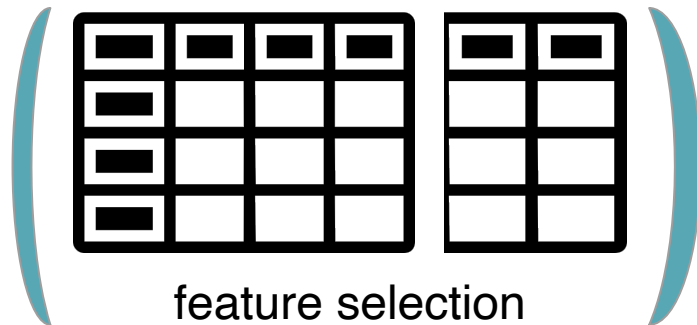
# THE MAIN TYPES OF MACHINE LEARNING



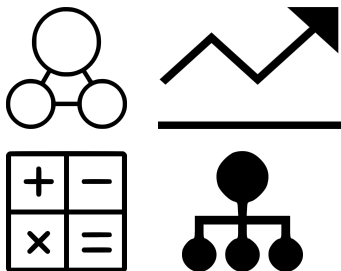
# The usual ingredients in the ML recipe



data  
partitioning



feature selection



model selection



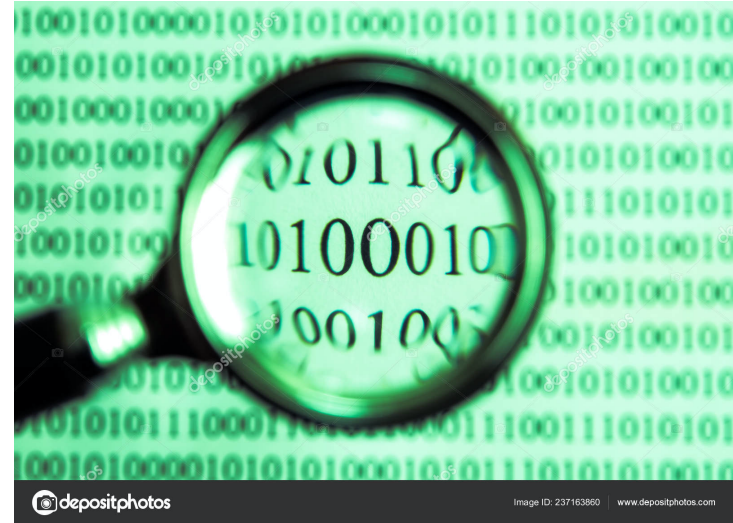
model assessment

Drowning in data?



Image borrowed from  
<https://medium.com/@tmaoconnor/marketing-is-drowning-in-data-yet-its-starving-for-answers-f15ee890a732>

Limited data?



depositphotos

Image ID: 237163860 www.depositphotos.com

Data partitioning:  
Evaluating how well you will  
generalize to new data with huge  
amounts of data

All the data

|   |   |   |   |
|---|---|---|---|
| ■ | ■ | ■ | ■ |
| ■ |   |   |   |
| ■ |   |   |   |
| ■ |   |   |   |

training set

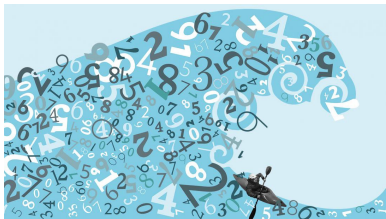
|   |  |  |  |
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| ■ |  |  |  |

Data used to build  
your model

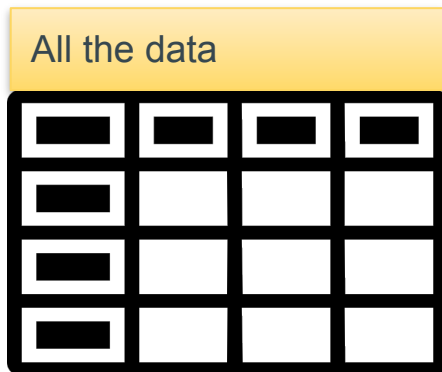
test set

|   |  |  |  |
|---|--|--|--|
| ■ |  |  |  |
| ■ |  |  |  |

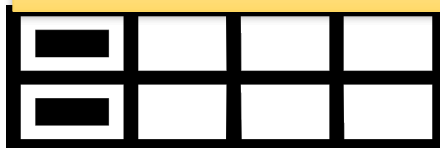
Used to assess how good  
the final model is  
(Generalization)



Data partitioning:  
Finding the best version of your  
model through validation



training set



Data used to build  
your model

validation set



Fine-tune the model to  
increase prediction  
accuracy  
(Hyper-parameter tuning,  
feature selection, model  
selection)

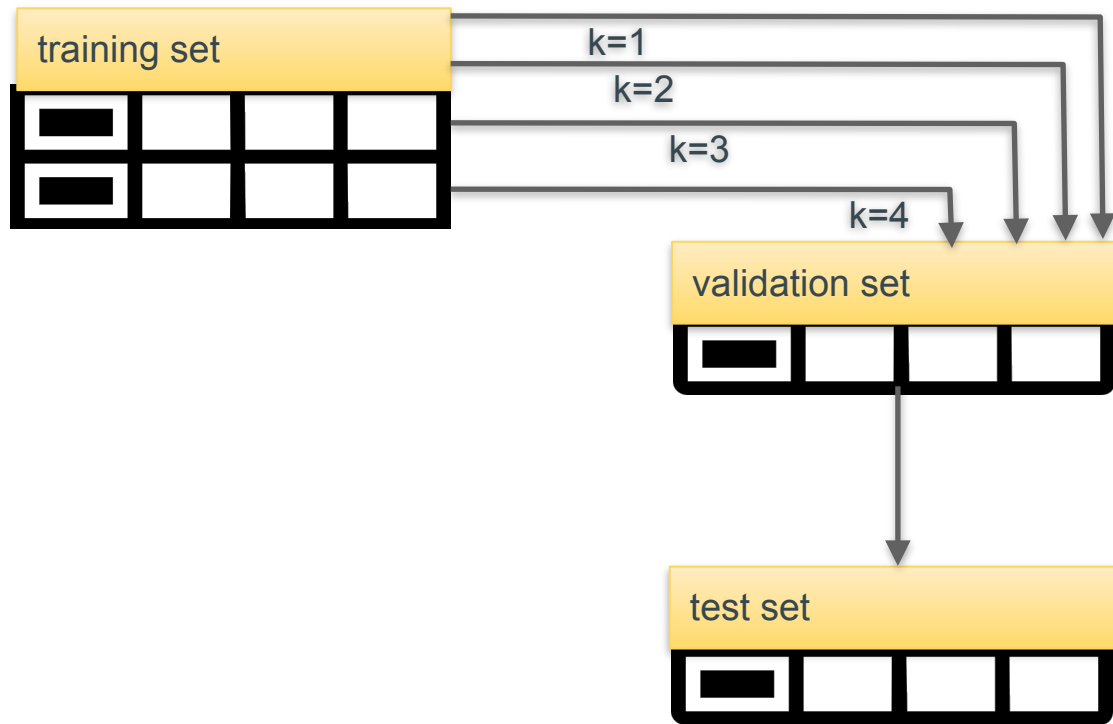
test set



Used to assess how good  
the final model is  
(Generalization)

# Selecting a model

Imagine we are using a ML model with a hyper parameter  $k$  taking values from 1 to 4

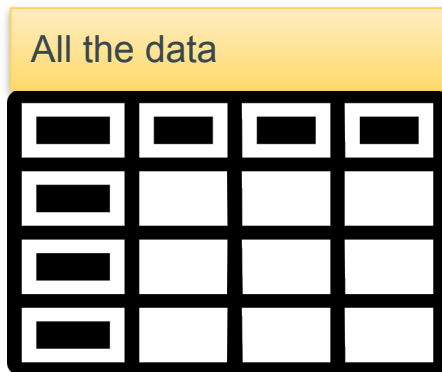


Pick the model that performs best on validation data...

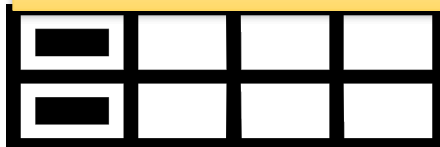
Now retrain that  $k$ -value on a combo of training set + validation set

And evaluate its future performance level on the test set

Data partitioning:  
Finding the best version of your  
model through validation



training set



Typically biggest  
chunk  
50 - 80%

validation set



Smaller  
10 - 25%

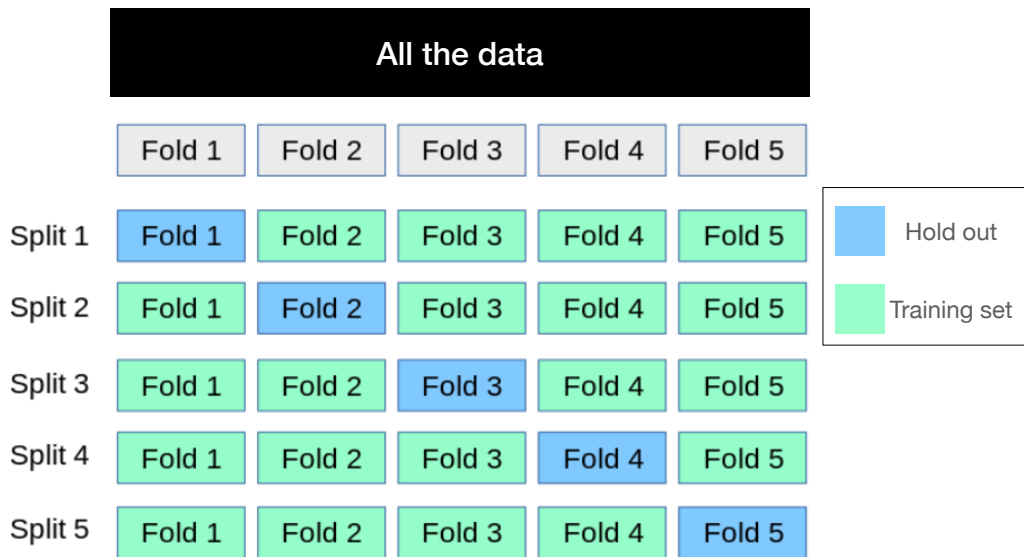
test set



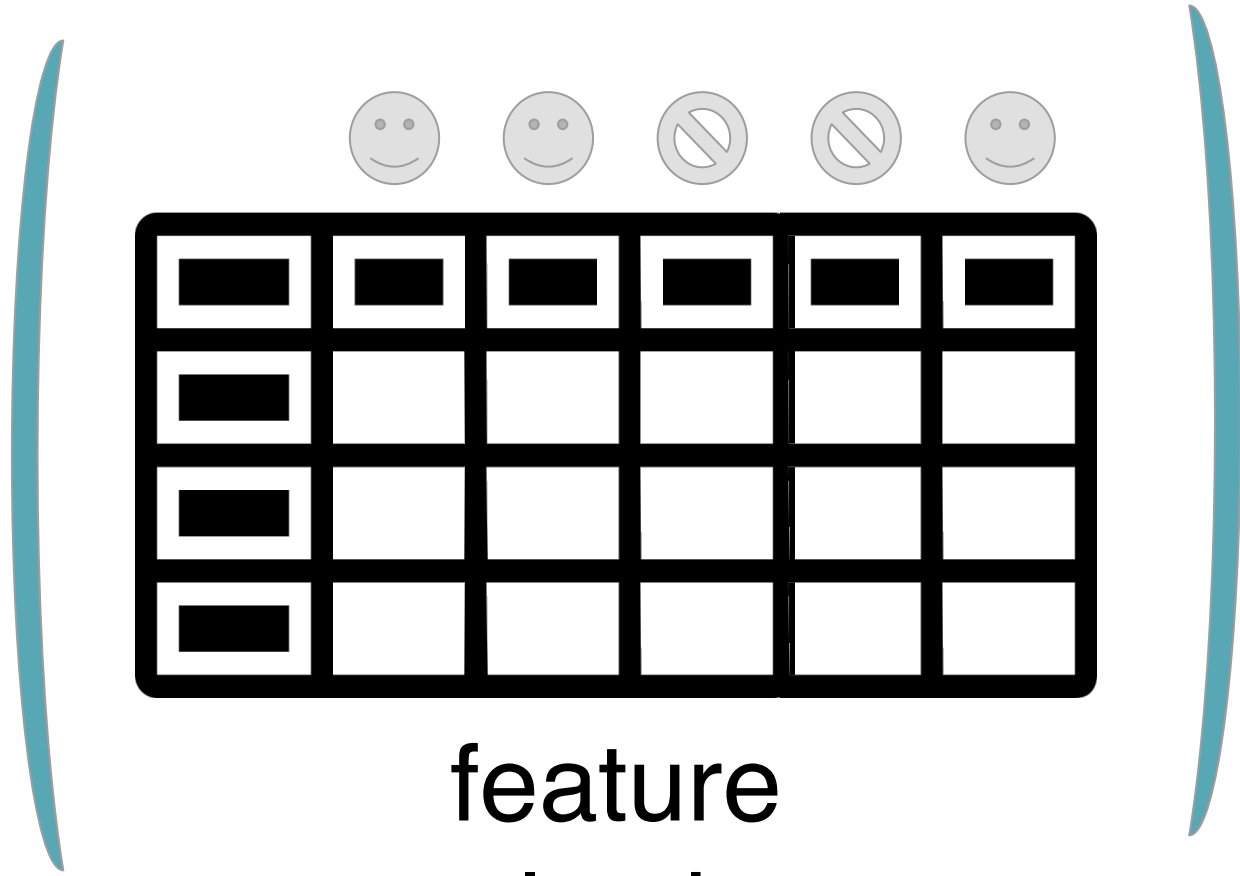
Smaller  
10 - 25%

# When you don't have huge data

## Cross validation

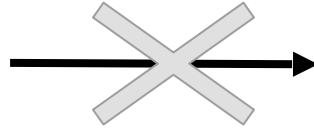
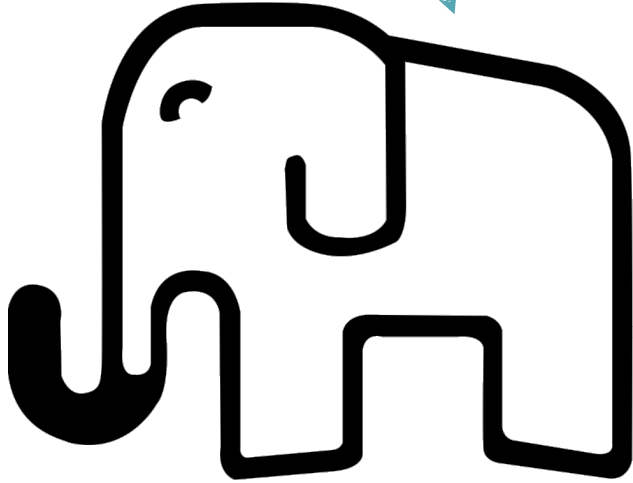


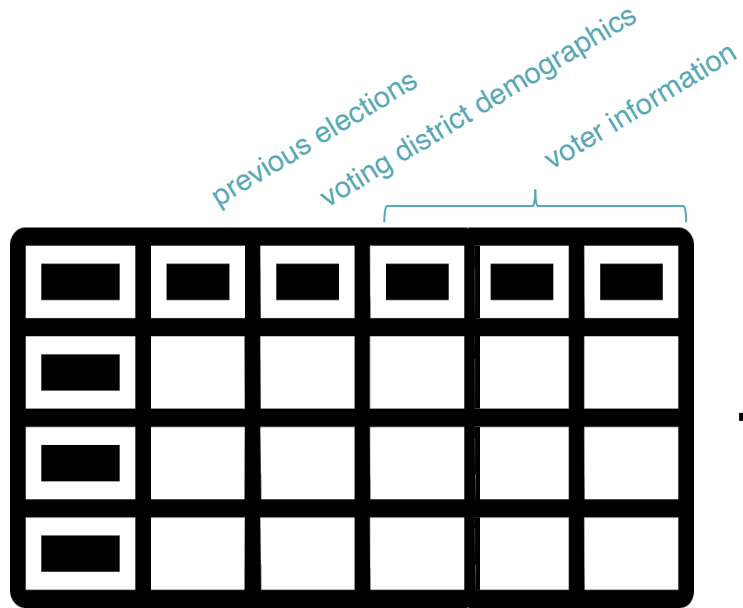
**Use the mean of the hold out set performances to estimate either test or validation error**



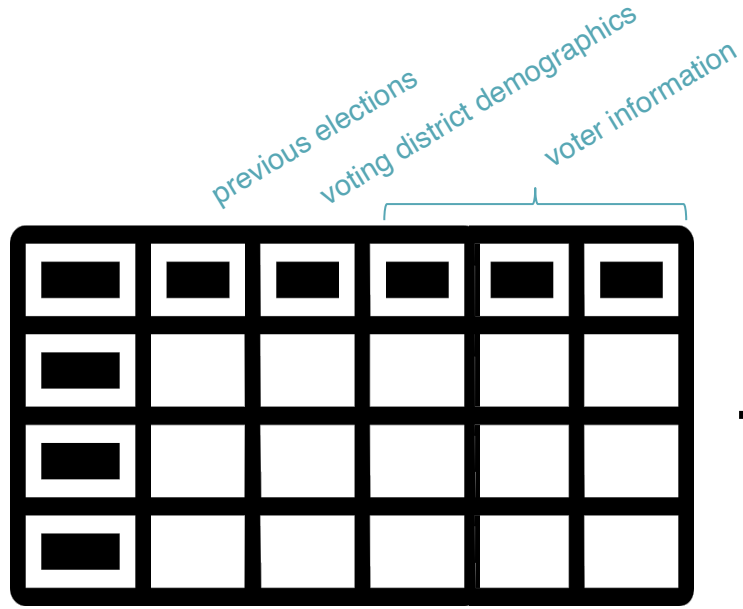
feature  
selection

elephant height data  
are likely not predictive  
of US elections



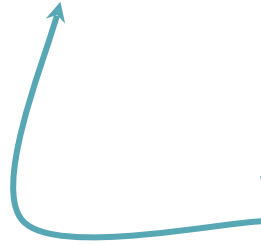


these data are likely  
predictive of US  
election outcomes



**feature selection** determines which variables are most predictive and includes them in the model

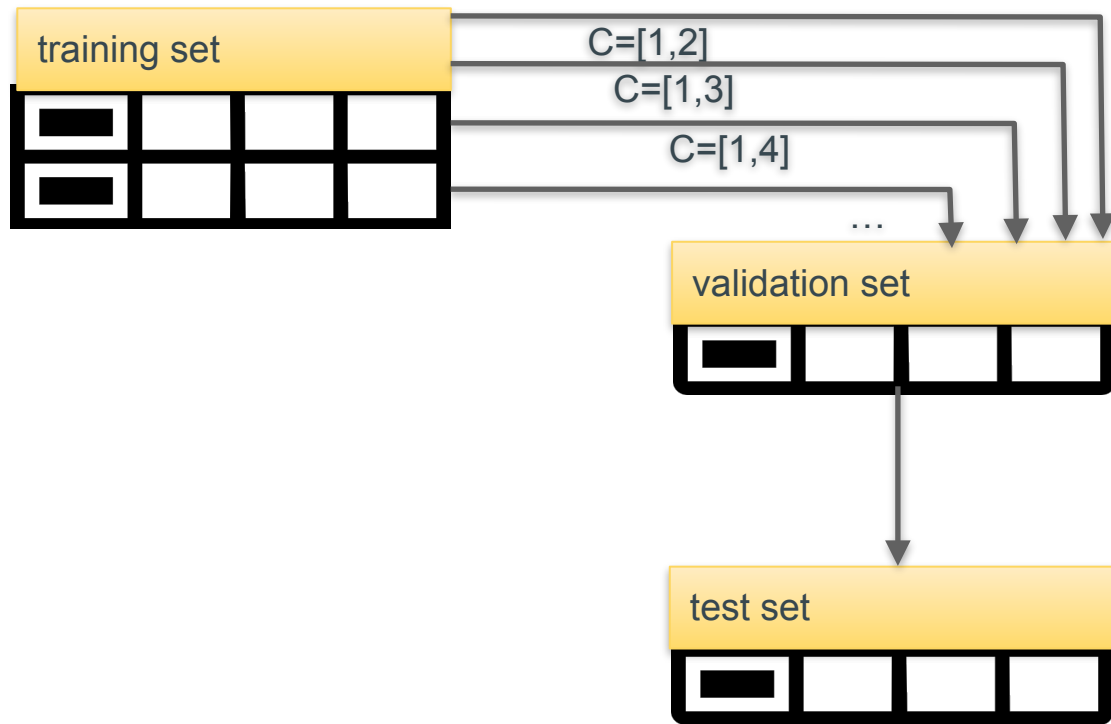
|   |   |   |   |   |   |
|---|---|---|---|---|---|
| ■ | ■ | ■ | ■ | ■ | ■ |
| ■ |   |   |   |   |   |
| ■ |   |   |   |   |   |
| ■ |   |   |   |   |   |



variables that can be used for accurate  
prediction exploit the relationship  
between the variables but do NOT  
mean that one causes the other

# Selecting a model

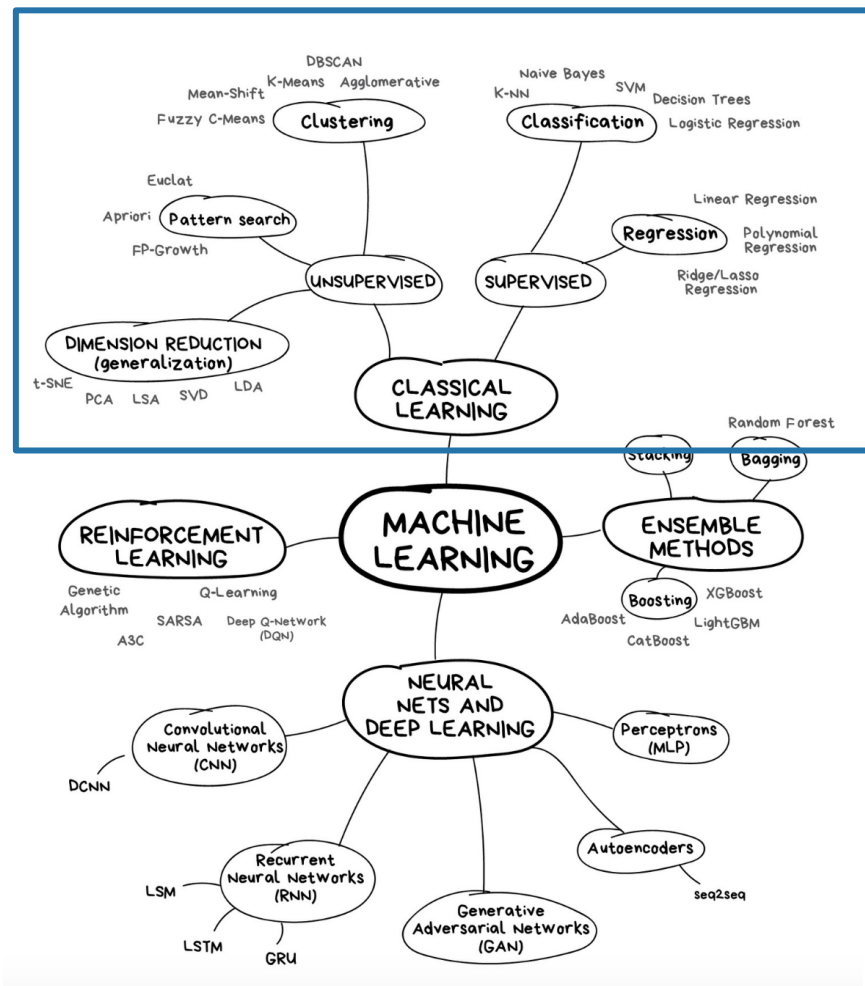
Feature selecting an ML model with different subsets of the data columns,  $C$



Pick the model that performs best on validation data...

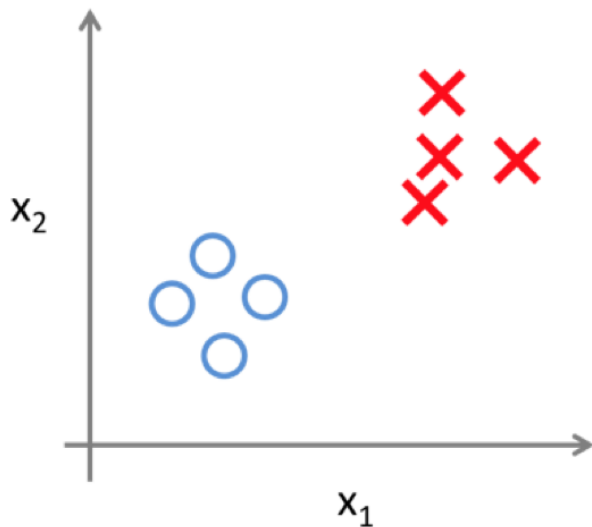
Now retrain that column subset on a combo of training set + validation set

And evaluate its future performance level on the test set



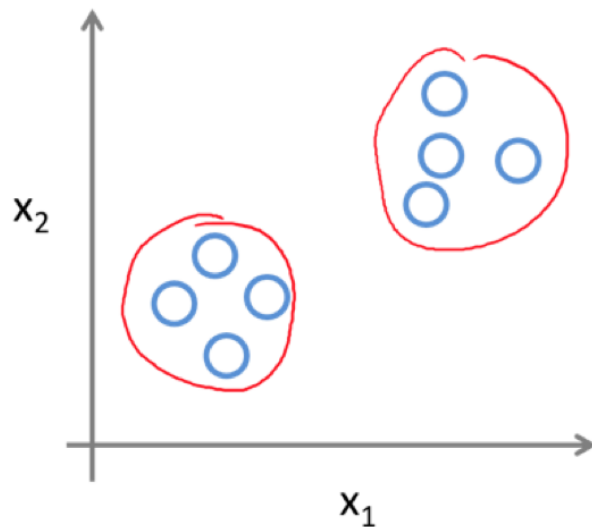
# Two modes of machine learning

## Supervised Learning



You give examples, the computer learns from them

## Unsupervised Learning



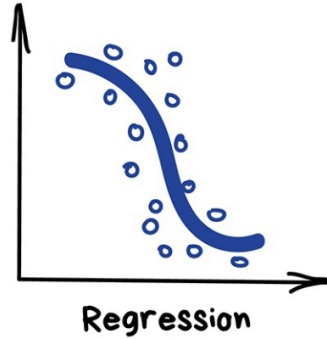
The computer determines how to classify based on properties within the data

## Regression

"Draw a line through these dots. Yep, that's the machine learning"

Today this is used for:

- Stock price forecasts
- Demand and sales volume analysis
- Medical diagnosis
- Any number-time correlations



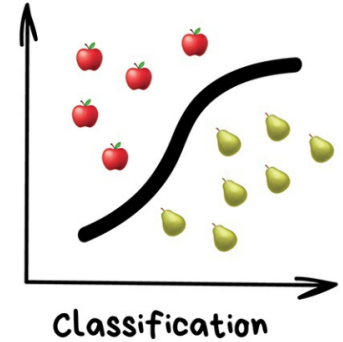
continuous variable prediction

## Classification

"Splits objects based at one of the attributes known beforehand. Separate socks by based on color, documents based on language, music by genre"

Today used for:

- Spam filtering
- Language detection
- A search of similar documents
- Sentiment analysis
- Recognition of handwritten characters and numbers
- Fraud detection

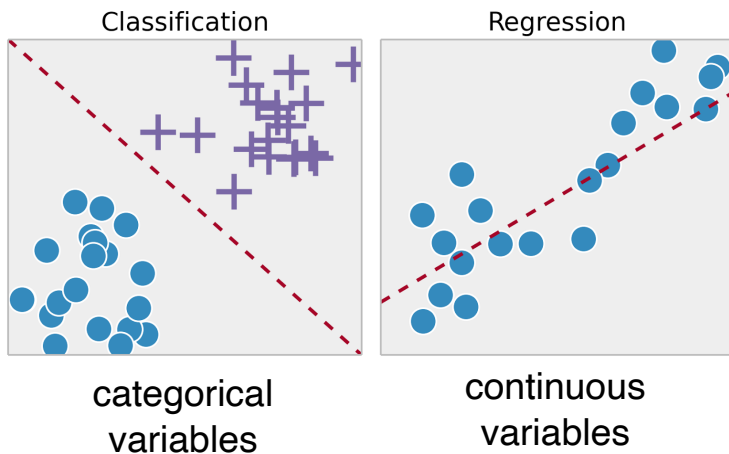


Popular algorithms: Naive Bayes, Decision Tree, Logistic Regression, K-Nearest Neighbours, Support Vector Machine

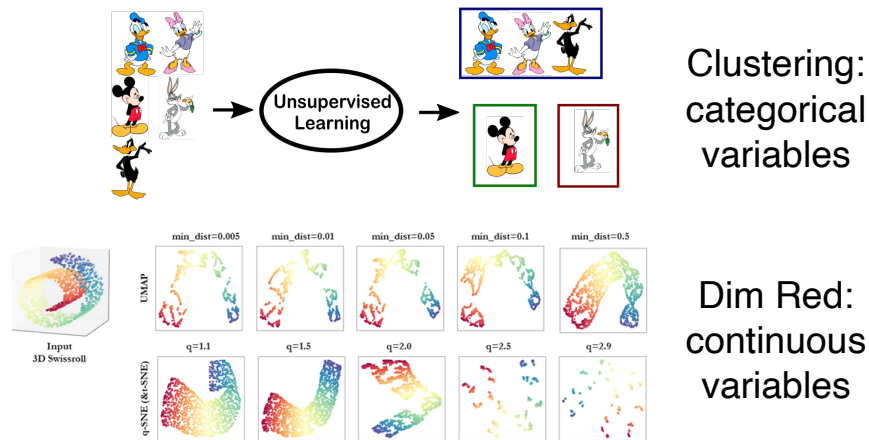
categorical variable prediction

# Approaches to machine learning

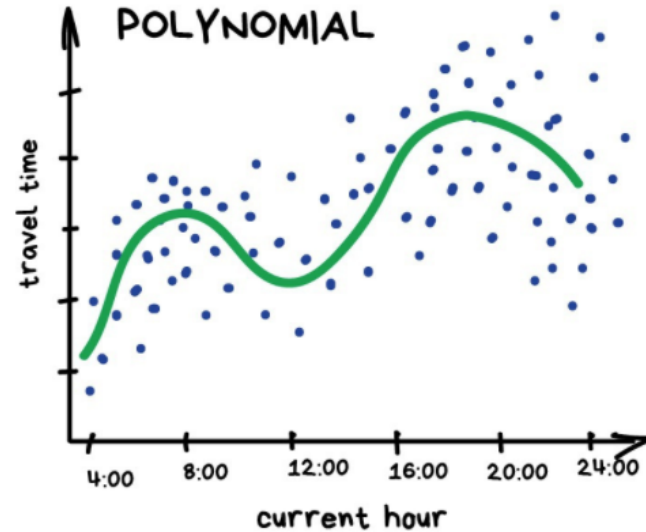
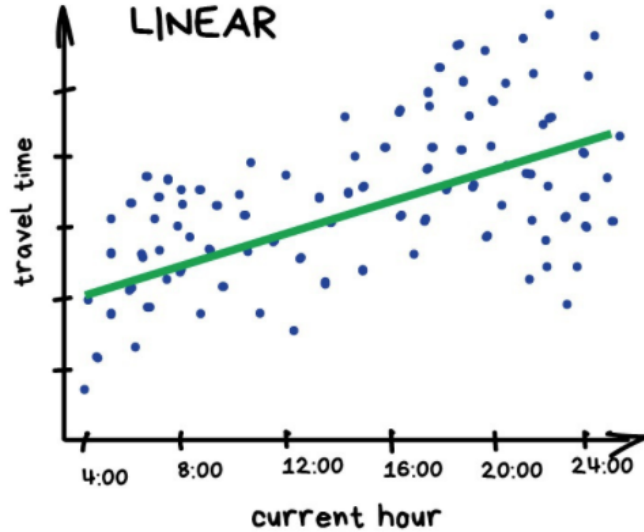
## Supervised Learning



## Unsupervised Learning



## PREDICT TRAFFIC JAMS



<https://forms.gle/HBiDbxR2oRBYxcUh7>

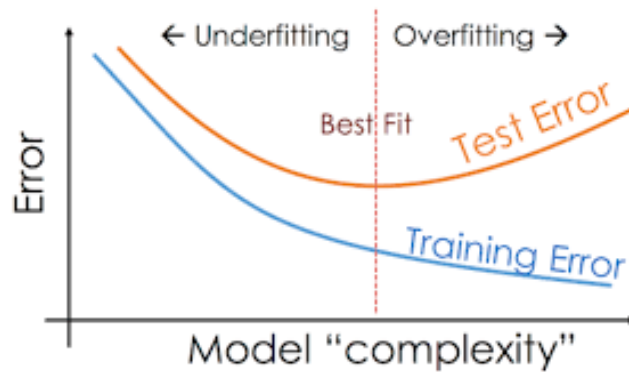
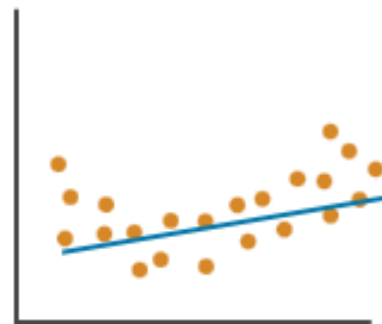
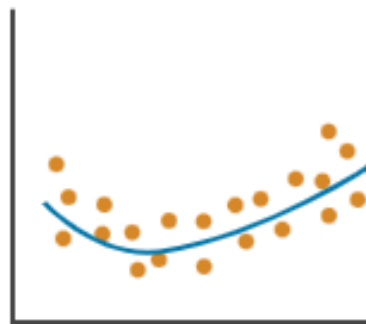
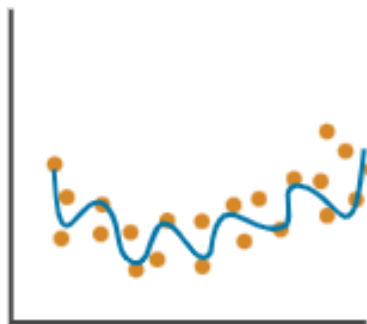


Overfit

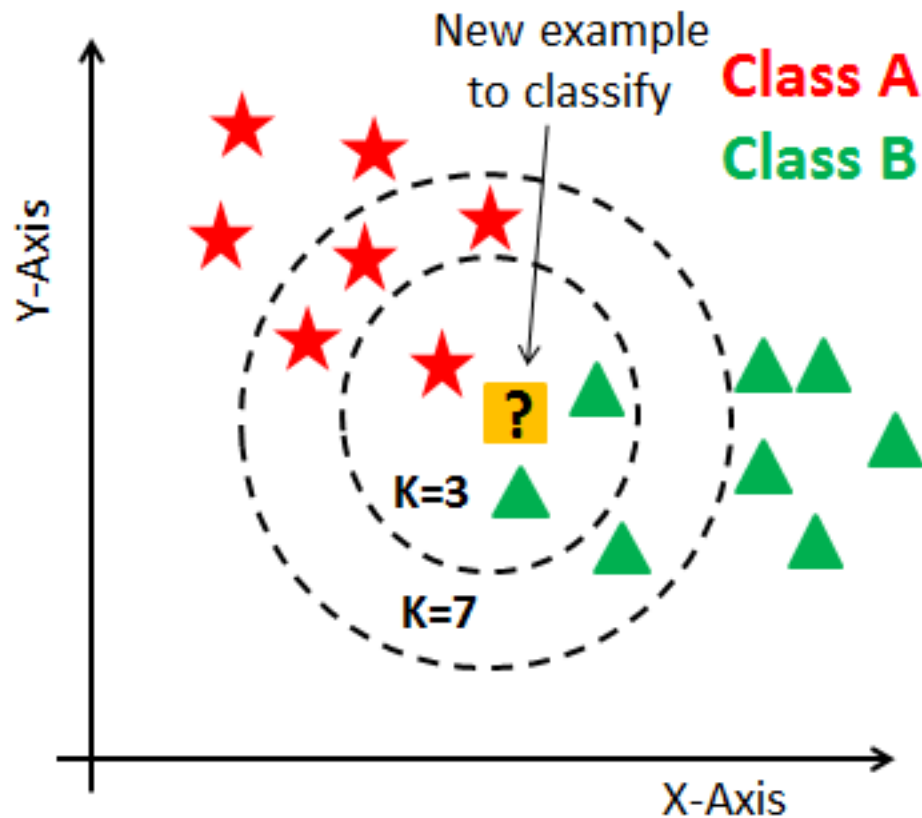
Best fit?

Underfit

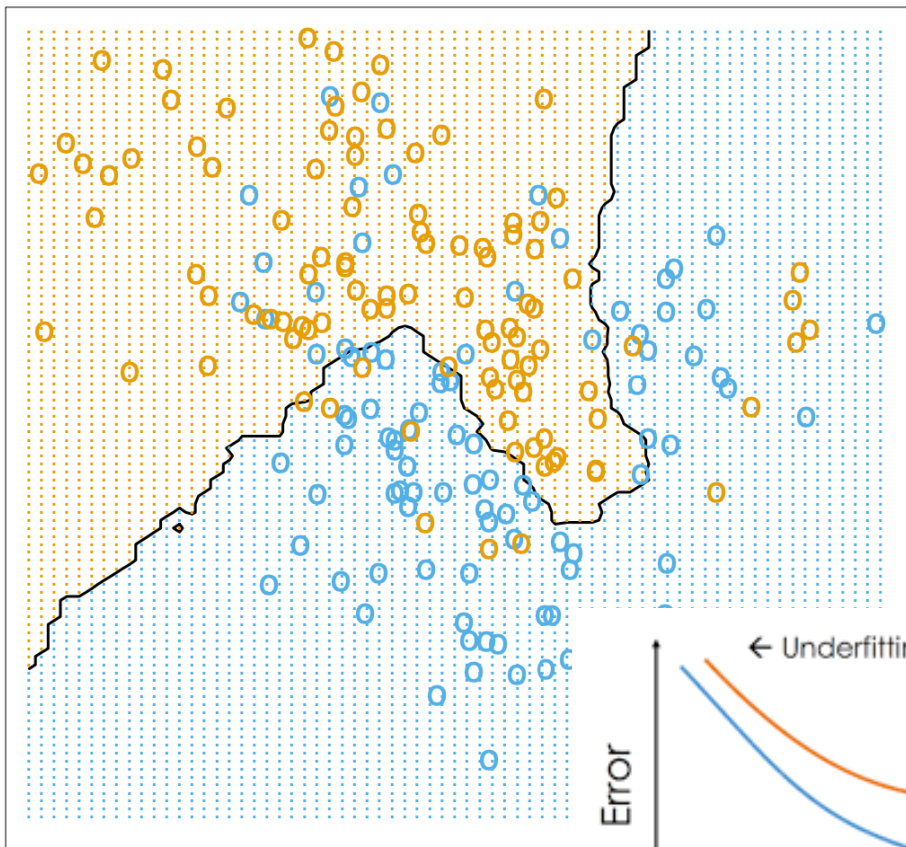
Regression



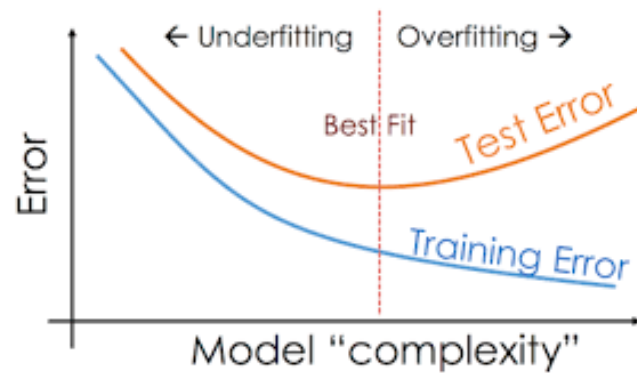
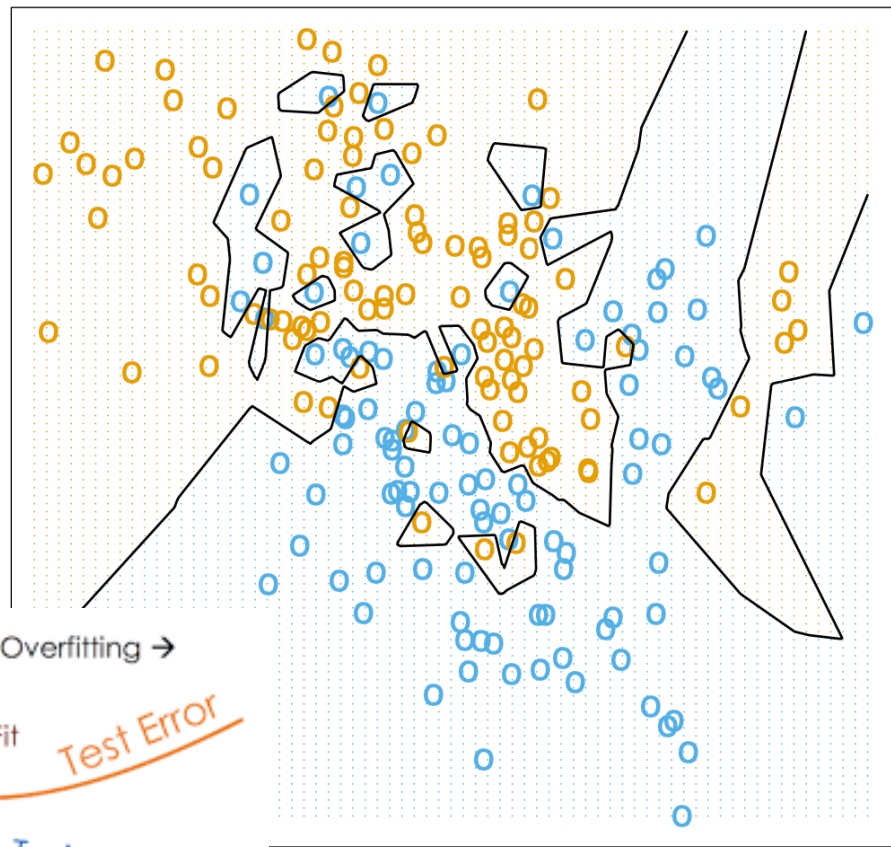
# K-nearest neighbors



15-Nearest Neighbor Classifier

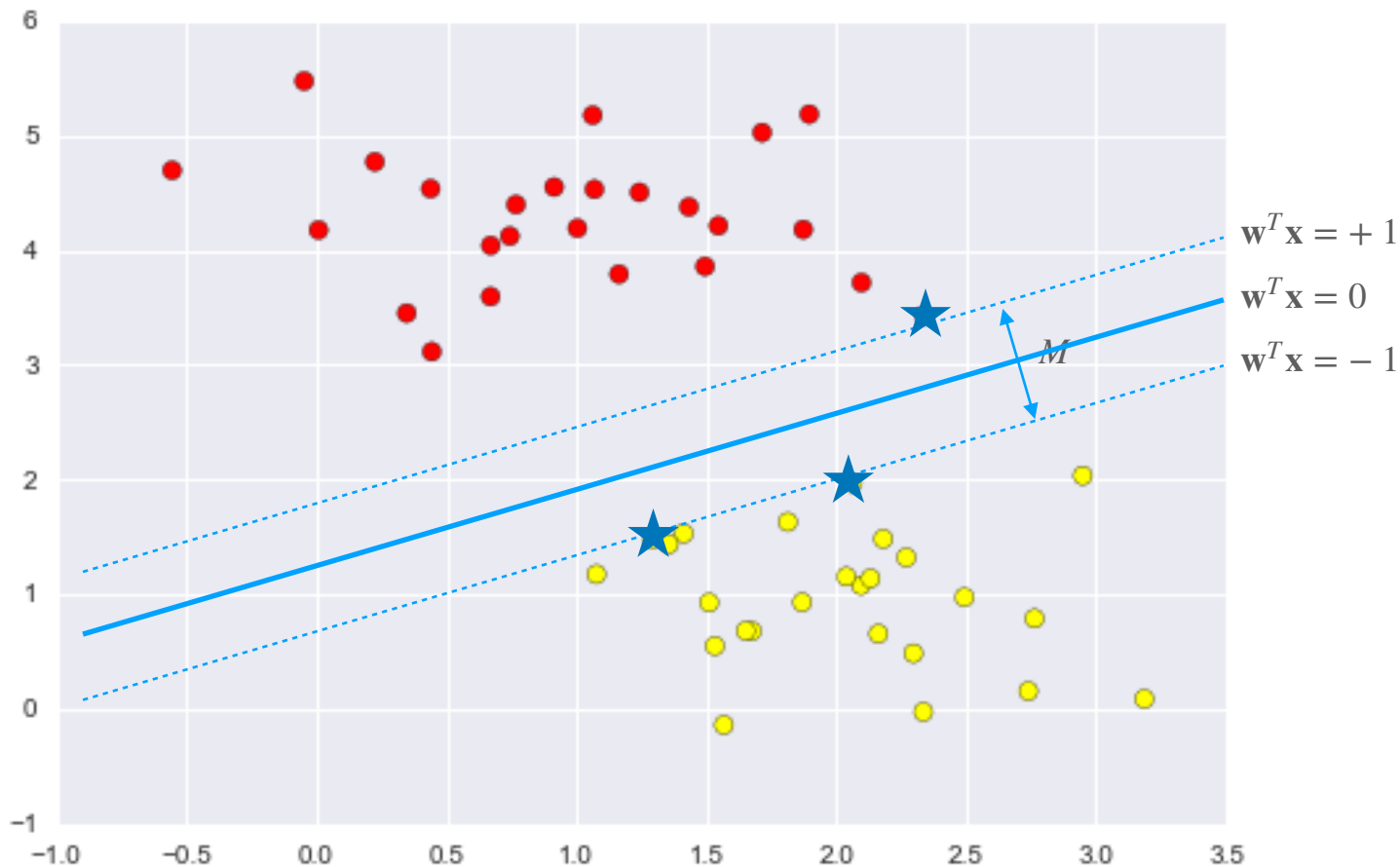


1-Nearest Neighbor Classifier



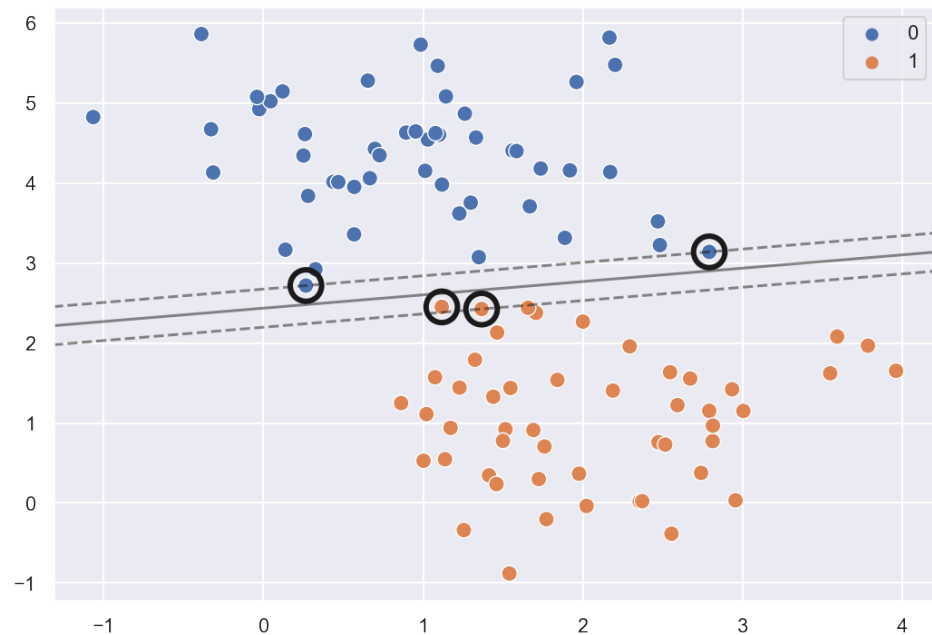
## Support Vector Machines:

Find the support  
data points that  
maximize the  
margin  $M$



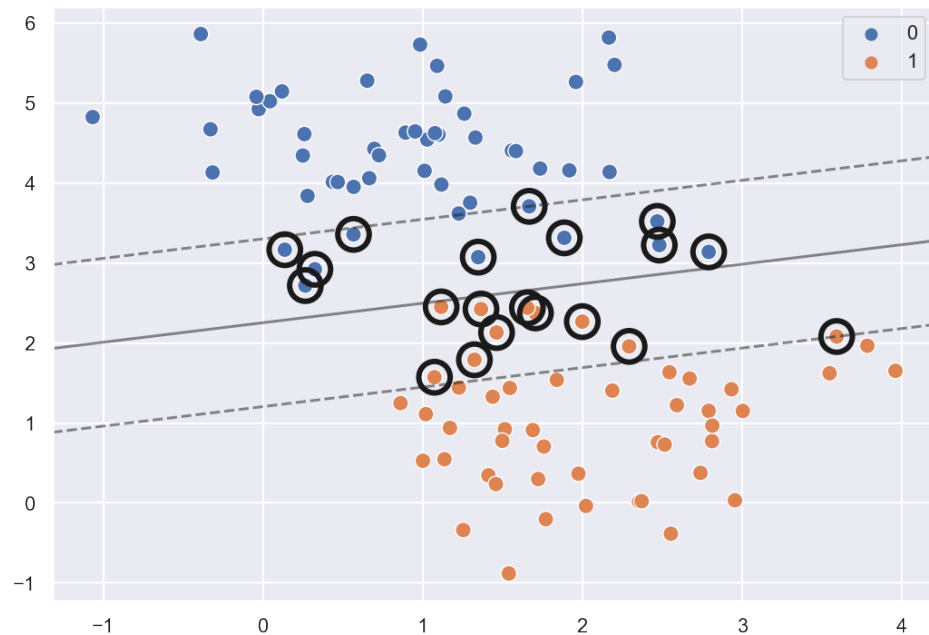
## Hard margin

$C = 10.0$



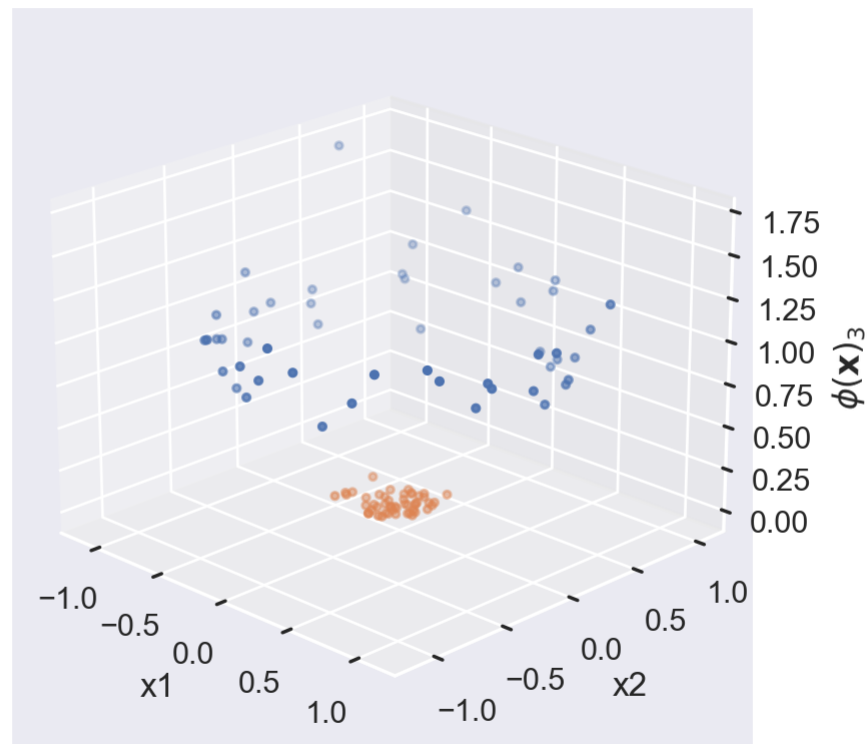
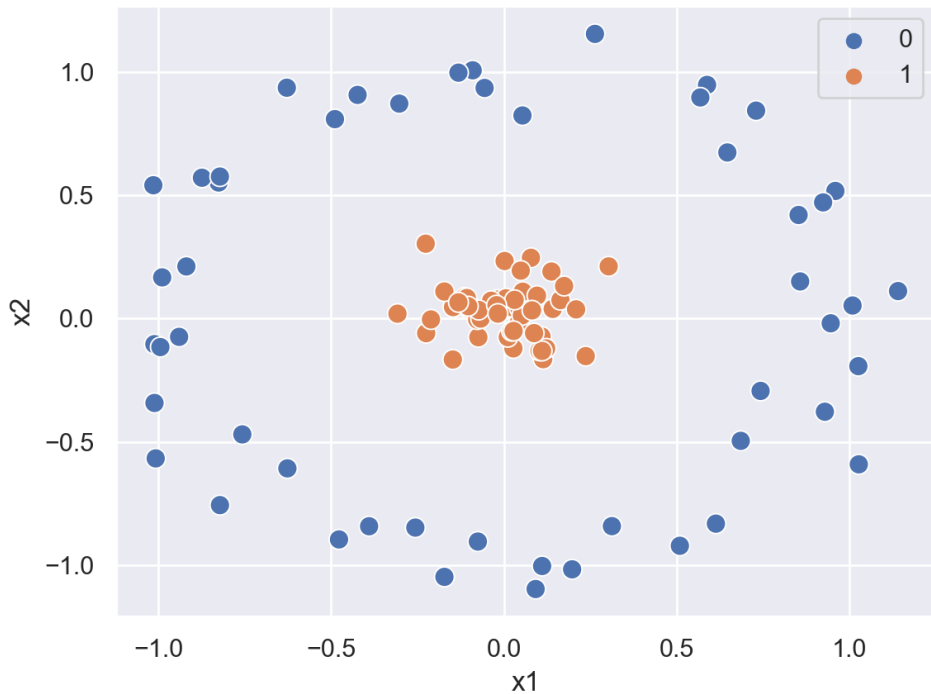
## Soft margin

$C = 0.1$

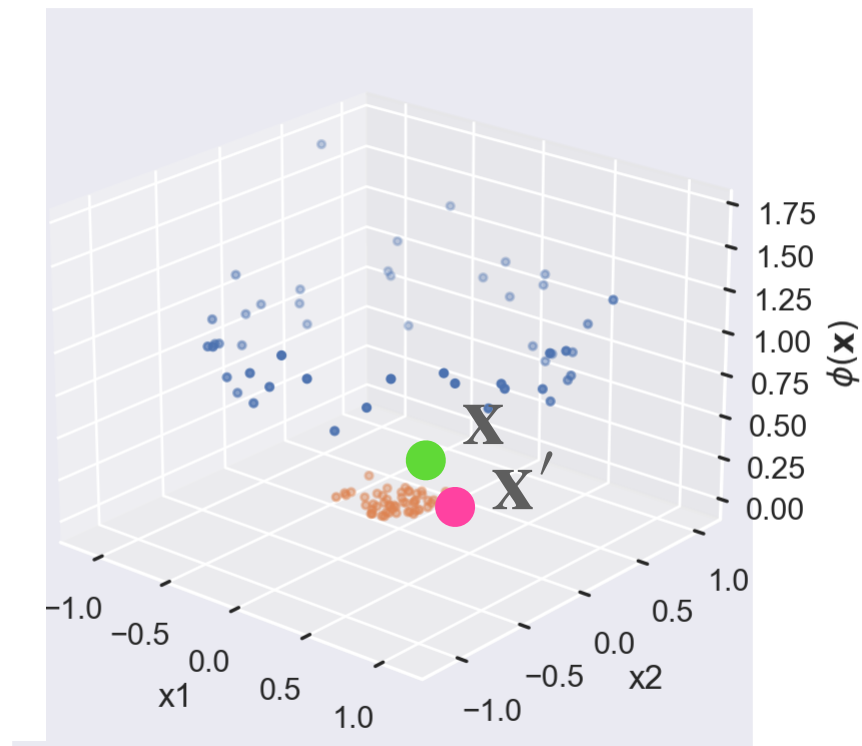
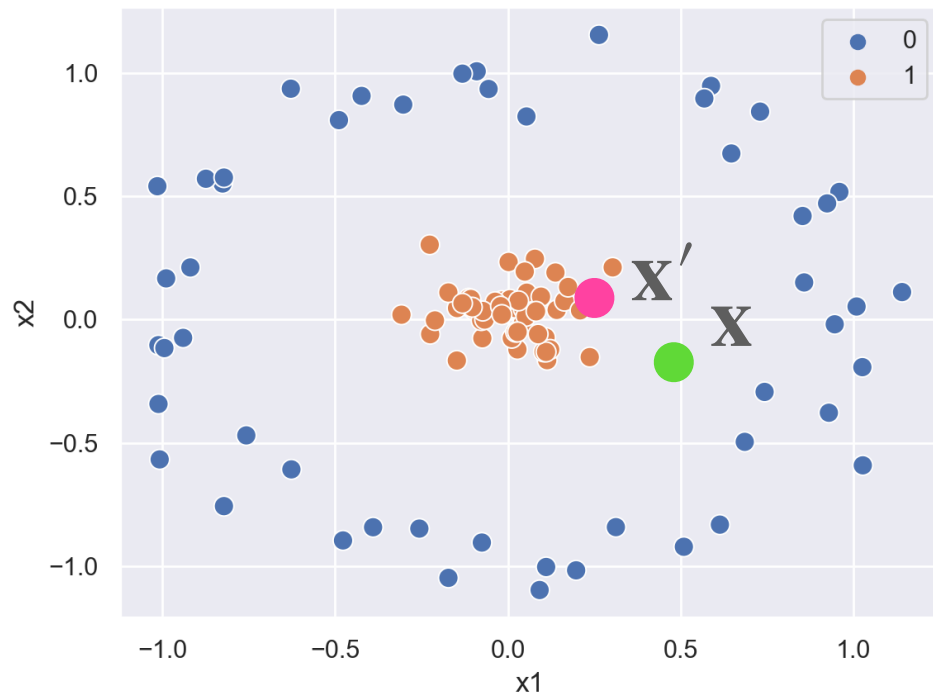


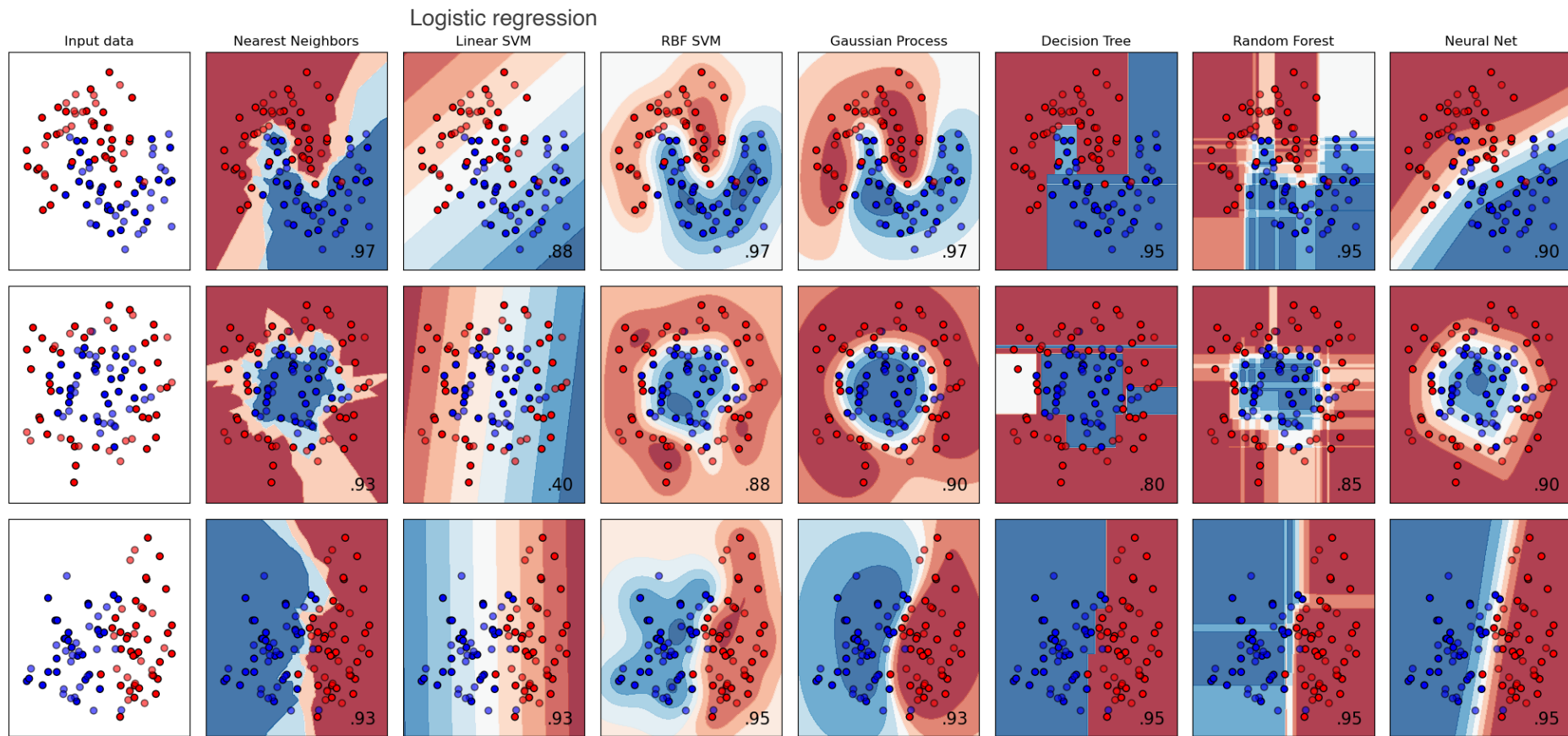
Find:  $\arg \min_{\mathbf{w}} \frac{1}{2} \|\mathbf{w}\|^2 + C \times \sum_{i=1}^n (1 - y_i \times (\mathbf{w}^T \mathbf{x}_i + b))_+$

$$\phi(\mathbf{x}) = [x_1, x_2, x_1 \cdot x_2 + x_1^2 \cdot x_2^2]$$



More generally this is the polynomial kernel





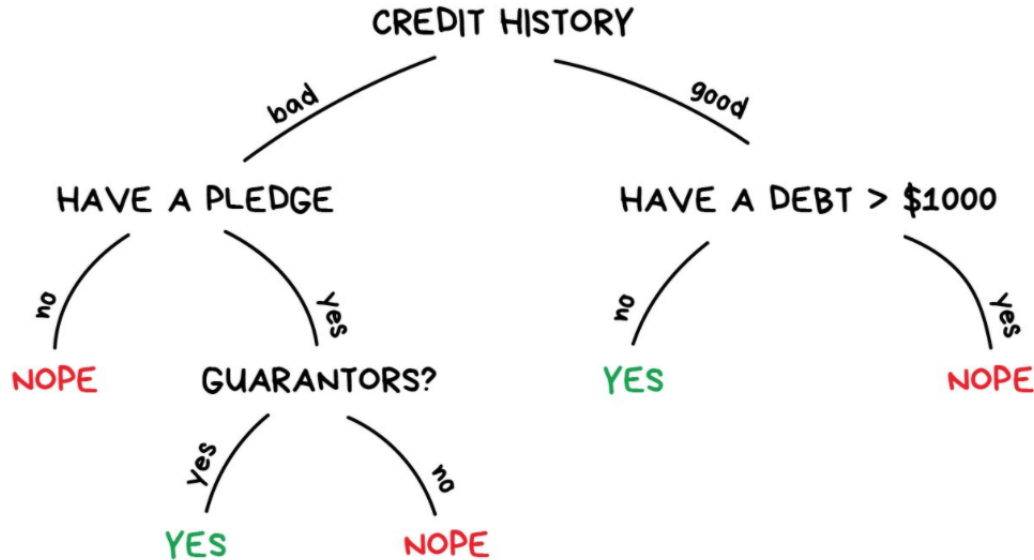
[https://scikit-learn.org/stable/auto\\_examples/classification/plot\\_classifier\\_comparison.html](https://scikit-learn.org/stable/auto_examples/classification/plot_classifier_comparison.html)

# Linear vs Non-linear classifiers

- LINEAR - Lower complexity, less overfitting, less power
  - Logistic regression
  - “Linear kernel” SVM
- NON-LINEAR - Higher complexity, more overfitting, more power
  - K-nearest Neighbors
  - “RBF kernel” SVM

Classification  
using  
a decision tree

## GIVE A LOAN?

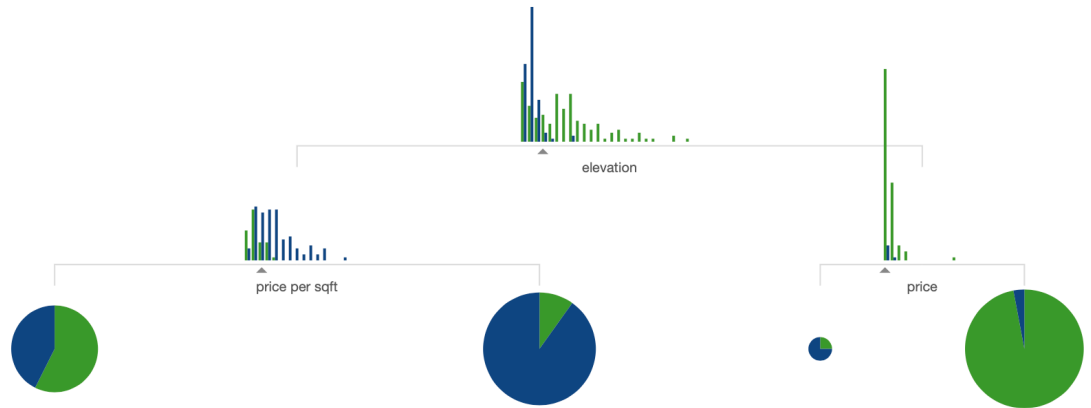


Is this server gone now????

## Growing a tree

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Additional forks will add new information that can increase a tree's **prediction accuracy**.



<http://www.r2d3.us/visual-intro-to-machine-learning-part-1/>